手寫作業 F44134057 [illegible]

11. Let A and B be 2×2 matrices.

(a) Does $\det(A+B) = \det(A) + \det(B)$?

(b) Does $\det(AB) = \det(A)\det(B)$?

(c) Does $\det(AB) = \det(BA)$?

(a) take a easy example

let $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, $A+B = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$

$\det(A) = 1 \times 1 = 1$, $\det(B) = 2 \times 3 = 6$, $\det(A+B) = 12$

$1 + 6 \neq 12$

$\Rightarrow \det(A+B) \neq \det(A) + \det(B) \Rightarrow$ False ※

(b)

let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$, $AB = \begin{bmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{bmatrix}$

$\left.\begin{array}{l} \det(A) = ad - cb \\ \det(B) = eh - gf \end{array}\right\} \Rightarrow \det(A)\cdot\det(B) = adeh - adfg + bcgf - bceh$

$\det(A \cdot B) = (ae+bg)\cdot(cf+dh) - (af+bh)(ce+dg)$

$= \cancel{acef} + adeh + bcgf + \cancel{bdgh} - \cancel{acef} - adfg - bceh - \cancel{bdgh}$

$= adeh + bcgf - adfg - bceh$

By comparision,

$\det(A)\cdot\det(B) = \det(A \cdot B) \Rightarrow$ True ※

(c) $\det(A \cdot B) = \det(A)\cdot\det(B)$, $\det(A)$ and $\det(B) \in \mathbb{R}$, which has commutative property

For $\mathbb{R}$, $\det(A)\cdot\det(B) = \det(B)\cdot\det(A)$

$= \det(B \cdot A)$

In summary, $\det(A \cdot B) = \det(B \cdot A) \Rightarrow$ True ※

2. Let

$$A = \begin{bmatrix} 2 & 1 & 2 & 2 \\ -3 & 3 & 1 & 3 \\ 2 & 1 & -1 & -4 \\ 1 & -3 & 2 & 0 \end{bmatrix}$$

(a) Use the elimination method to evaluate det(A). 30

(b) Use the value of det(A) to evaluate 60

$$\begin{vmatrix} 2 & 1 & 2 & 2 \\ 2 & 1 & -1 & -4 \\ 1 & -3 & 2 & 0 \\ -3 & 3 & 1 & 3 \end{vmatrix} + \begin{vmatrix} 2 & 1 & 2 & 2 \\ -3 & 3 & 1 & 3 \\ -1 & 4 & 0 & -1 \\ -2 & 0 & 3 & 3 \end{vmatrix}$$

$-3 \times \frac{37}{27}$

(a)
$$\begin{vmatrix} 2 & 1 & 2 & 2 \\ -3 & 3 & 1 & 3 \\ 2 & 1 & -1 & -4 \\ 1 & -3 & 2 & 0 \end{vmatrix} = \begin{vmatrix} 2 & 1 & 2 & 2 \\ 0 & 9/2 & 4 & 6 \\ 0 & 0 & -3 & -6 \\ 0 & -7/2 & 1 & -1 \end{vmatrix} = \begin{vmatrix} 2 & 1 & 2 & 2 \\ 0 & 9/2 & 4 & 6 \\ 0 & 0 & -3 & -6 \\ 0 & 0 & \frac{37}{9} & \frac{11}{3} \end{vmatrix}$$

$$= \begin{vmatrix} 2 & 1 & 2 & 2 \\ 0 & 9/2 & 4 & 6 \\ 0 & 0 & -3 & -6 \\ 0 & 0 & 0 & -\frac{41}{9} \end{vmatrix} = 2 \times \frac{9}{2} \times -3 \times -\frac{41}{9} = 123$$

(b)
$$\begin{vmatrix} 2 & 1 & 2 & 2 \\ 2 & 1 & -1 & -4 \\ 1 & -3 & 2 & 0 \\ -3 & 3 & 1 & 3 \end{vmatrix} + \begin{vmatrix} 2 & 1 & 2 & 2 \\ -3 & 3 & 1 & 3 \\ -1 & 4 & 0 & -1 \\ -2 & 0 & 3 & 3 \end{vmatrix} \; r_2 \times -1$$

$$= -\begin{vmatrix} 2 & 1 & 2 & 2 \\ -3 & 3 & 1 & 3 \\ 1 & -3 & 2 & 0 \\ 2 & 1 & -1 & -4 \end{vmatrix} + \begin{vmatrix} 2 & 1 & 2 & 2 \\ -3 & 3 & 1 & 3 \\ 2 & 1 & -1 & -4 \\ -2 & 0 & 3 & 3 \end{vmatrix} \; r_2 \times -1$$

$$= \begin{vmatrix} 2 & 1 & 2 & 2 \\ -3 & 3 & 1 & 3 \\ 2 & 1 & -1 & -4 \\ 1 & -3 & 2 & 0 \end{vmatrix} + \begin{vmatrix} 2 & 1 & 2 & 2 \\ -3 & 3 & 1 & 3 \\ 2 & 1 & -1 & -4 \\ 1 & -3 & 2 & 0 \end{vmatrix} = 123 + 123 = 246 \text{※}$$

18. Let A be a $k \times k$ matrix and let B be an $(n-k) \times (n-k)$ matrix. Let

$$E = \begin{bmatrix} I_k & O \\ O & B \end{bmatrix}, \quad F = \begin{bmatrix} A & O \\ O & I_{n-k} \end{bmatrix},$$

$$C = \begin{bmatrix} A & O \\ O & B \end{bmatrix}$$

where I_k and I_{n-k} are the $k \times k$ and $(n-k) \times (n-k)$ identity matrices.

(a) Show that $\det(E) = \det(B)$.

(b) Show that $\det(F) = \det(A)$.

(c) Show that $\det(C) = \det(A)\det(B)$.

(a) For $E = \begin{bmatrix} I_k & 0 \\ 0 & B \end{bmatrix}$, we know row addition wouldn't change the determint value, so we can apply Gaussian elimination to this matrix.
Note Upper triangular form of B as $U(B)$, and we know $\det(B) = \det(U(B))$

$$\underset{n\times n}{\begin{vmatrix} I_k & 0 \\ 0 & B \end{vmatrix}} \longrightarrow \underset{n\times n}{\begin{vmatrix} I_k & 0 \\ 0 & U(B) \end{vmatrix}}$$

, to have $\det(E)$ we need to compute the product of diagonal entries, $\prod_{i=1}^{k} I_{ii} \cdot \prod_{j=1}^{n-k} U(B)_{jj}$,

which $\prod_{i=1}^{k} I_{ii} = 1$, $\prod_{j=1}^{n-k} U(B)_{jj} = \det(U(B)) = \det(B)$.

In result, $\det(E) = \prod_{i=1}^{k} I_{ii} \times \prod_{j=1}^{n-k} U(B)_{jj} = \det(B)$ ※

(b) $F = \begin{bmatrix} A & 0 \\ 0 & I_{n-k} \end{bmatrix}_{n\times n}$, apply the same idea to prove $Det(F) = Det(A)$

Note reduced row echelon form of A as $V(A)$, $(\det(A) = \det(V(A)))$, $\begin{vmatrix} A & 0 \\ 0 & I_{n-k} \end{vmatrix} = \begin{vmatrix} V(A) & 0 \\ 0 & I_{n-k} \end{vmatrix}$

$$\det(F) = \prod_{i=0}^{k} V(A)_{ii} \times \prod_{j=1}^{n-k} I_{jj} = \det(V(A)) \times 1 = \det(A)$$

$\Rightarrow \det(F) = \det(A)$ #

(c) $C = \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}_{n\times n}$, with same technique, $\begin{vmatrix} A & 0 \\ 0 & B \end{vmatrix} \longrightarrow \begin{vmatrix} V(A) & 0 \\ 0 & V(B) \end{vmatrix}$

$$\det(C) = \prod_{i=1}^{k} V(A)_{ii} \times \prod_{j=1}^{n-k} V(B)_{jj} = \det(V(A)) \cdot \det(V(B)) = \det(A) \cdot \det(B)$$

$\Rightarrow \det(C) = \det(A) \cdot \det(B)$ #

2. Use Cramer's rule to solve each of the following systems:

(a) $2x_1 + 3x_2 = 2$
$8x_1 - 6x_2 = 2$

(b) $6x_1 + 3x_2 = 5$
$5x_1 + 7x_2 = 12$

(c) $4x_1 - 2x_3 = 26$
$7x_1 + 7x_2 + 2x_3 = -4$
$3x_1 - 2x_2 + 1x_3 = -3$

(d) $2x_1 - 2x_2 + 3x_3 = 0$
$x_1 + 2x_2 + 3x_3 = 8$
$-2x_1 + 4x_2 + x_3 = 6$

(e) $-x_1 + 2x_2 + x_3 + x_4 = 9$
$2x_1 + x_2 - 2x_3 + x_4 = 3$
$3x_1 + 3x_2 + x_3 = 6$
$2x_2 + x_3 - x_4 = 2$

(a) $\begin{bmatrix} 2 & 3 \\ 8 & -6 \end{bmatrix}\begin{bmatrix} | \\ x \\ | \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

$x_1 = \dfrac{\det(B_1)}{\det(A)} = \dfrac{\begin{vmatrix} 2 & 3 \\ 2 & -6 \end{vmatrix}}{\begin{vmatrix} 2 & 3 \\ 8 & -6 \end{vmatrix}} = \dfrac{-12-6}{-12-24} = \dfrac{-18}{-36} = \dfrac{1}{2}$

$x_2 = \dfrac{\det(B_2)}{\det(A)} = \dfrac{\begin{vmatrix} 2 & 2 \\ 8 & 2 \end{vmatrix}}{\begin{vmatrix} 2 & 3 \\ 8 & -6 \end{vmatrix}} = \dfrac{4-16}{-12-24} = \dfrac{-12}{-36} = \dfrac{1}{3}$

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{3} \end{bmatrix}$ ※

(b) $\begin{bmatrix} 6 & 3 \\ 5 & 7 \end{bmatrix}\begin{bmatrix} | \\ x \\ | \end{bmatrix} = \begin{bmatrix} 5 \\ 12 \end{bmatrix}$

$x_1 = \dfrac{\det(B_1)}{\det(A)} = \dfrac{\begin{vmatrix} 5 & 3 \\ 12 & 7 \end{vmatrix}}{\begin{vmatrix} 6 & 3 \\ 5 & 7 \end{vmatrix}} = \dfrac{35-36}{42-15} = \dfrac{-1}{27}$

$x_2 = \dfrac{\begin{vmatrix} 6 & 5 \\ 5 & 12 \end{vmatrix}}{\begin{vmatrix} 6 & 3 \\ 5 & 7 \end{vmatrix}} = \dfrac{72-25}{42-15} = \dfrac{47}{27}$

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \frac{-1}{27} \\ \frac{47}{27} \end{bmatrix}$ ※

(c) $\begin{bmatrix} 4 & 0 & -2 \\ 7 & 7 & 2 \\ 3 & -2 & 1 \end{bmatrix}\begin{bmatrix} | \\ x \\ | \end{bmatrix} = \begin{bmatrix} 26 \\ -4 \\ -3 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 10 & -4 & 0 \\ 1 & 11 & 0 \\ 3 & -2 & 1 \end{bmatrix}\begin{bmatrix} | \\ x \\ | \end{bmatrix} = \begin{bmatrix} 20 \\ 2 \\ -3 \end{bmatrix}$

$x_1 = \dfrac{\begin{vmatrix} 20 & -4 & 0 \\ 2 & 11 & 0 \\ -3 & -2 & 1 \end{vmatrix}}{\begin{vmatrix} 10 & -4 & 0 \\ 1 & 11 & 0 \\ 3 & -2 & 1 \end{vmatrix}} = \dfrac{\begin{vmatrix} 20 & -4 \\ 2 & 11 \end{vmatrix}}{\begin{vmatrix} 10 & -4 \\ 1 & 11 \end{vmatrix}} = \dfrac{228}{114} = 2$

$x_2 = \dfrac{\begin{vmatrix} 10 & 20 & 0 \\ 1 & 2 & 0 \\ 3 & -3 & 1 \end{vmatrix}}{\begin{vmatrix} A \end{vmatrix}} = \dfrac{\begin{vmatrix} 10 & 20 \\ 1 & 2 \end{vmatrix}}{114} = 0$

$$x_3 = \frac{\begin{vmatrix} 10 & -4 & 20 \\ 1 & 11 & 2 \\ 3 & -2 & -3 \end{vmatrix}}{|A|}$$

$$= \frac{[\overset{-330}{10\times 11\times(-3)} + \overset{-24}{2\times 3\times(-4)} + \overset{-40}{20\times 1\times(-2)}] - [\overset{660}{20\times 11\times 3} + \overset{-40}{10\times 2\times(-2)} + \overset{12}{1\times(-4)\times 3}]}{114}$$

$$= \frac{\overset{-1026}{\cancel{-394-632}}}{114} = -9$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ -9 \end{bmatrix} \#$$

(d)
$$\begin{bmatrix} 2 & -2 & 3 \\ 1 & 2 & 3 \\ -2 & 4 & 1 \end{bmatrix} \begin{bmatrix} \\ X \\ \\ \end{bmatrix} = \begin{bmatrix} 0 \\ 8 \\ 6 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & -2 & 3 \\ 3 & 0 & 6 \\ 2 & 0 & 7 \end{bmatrix} \begin{bmatrix} \\ X \\ \\ \end{bmatrix} = \begin{bmatrix} 0 \\ 8 \\ 6 \end{bmatrix}$$

$$x_1 = \frac{\begin{vmatrix} 0 & -2 & 3 \\ 8 & 0 & 6 \\ 6 & 0 & 7 \end{vmatrix}}{\begin{vmatrix} 2 & -2 & 3 \\ 3 & 0 & 6 \\ 2 & 0 & 7 \end{vmatrix}} = \frac{\cancel{-(-2)}\begin{vmatrix} 8 & 6 \\ 6 & 7 \end{vmatrix}}{\cancel{-(-2)}\begin{vmatrix} 3 & 6 \\ 2 & 7 \end{vmatrix}} = \frac{56-36}{21-12} = \frac{20}{9}$$

$$x_2 = \frac{\begin{vmatrix} 2 & 0 & 3 \\ 3 & 8 & 6 \\ 2 & 6 & 7 \end{vmatrix}}{|A|} = \frac{8\cdot\begin{vmatrix} 2 & 3 \\ 2 & 7 \end{vmatrix} - 6\begin{vmatrix} 2 & 3 \\ 3 & 6 \end{vmatrix}}{2\times 9} = \frac{8\times 8 - 6\times 3}{2\times 9} = \frac{23}{9}$$

$$x_3 = \frac{\begin{vmatrix} 2 & -2 & 0 \\ 3 & 0 & 8 \\ 2 & 0 & 6 \end{vmatrix}}{|A|} = \frac{2\times\begin{vmatrix} 3 & 8 \\ 2 & 6 \end{vmatrix}}{2\times 9} = \frac{2\times(\overset{2}{\cancel{18-16}})}{2\times 9} = \frac{2}{9}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 20/9 \\ 23/9 \\ 2/9 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 2 & 1 & 1 \\ 2 & 1 & -2 & 1 \\ 3 & 3 & 1 & 0 \\ 0 & 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} \\ X \\ \\ \end{bmatrix} = \begin{bmatrix} 9 \\ 3 \\ 6 \\ 2 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 2 & 1 & 1 \\ 0 & 5 & 0 & 3 \\ 0 & 9 & 4 & 3 \\ 0 & 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} \\ X \\ \\ \end{bmatrix} = \begin{bmatrix} 9 \\ 21 \\ 33 \\ 2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} -1 & 2 & 1 & 1 \\ 3 & -1 & -3 & 0 \\ 3 & 3 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \\ X \\ \\ \end{bmatrix} = \begin{bmatrix} 9 \\ -6 \\ 6 \\ -7 \end{bmatrix}$$

$$x_1 = \frac{\begin{vmatrix} 9 & 2 & 1 & 1 \\ -6 & -1 & -3 & 0 \\ 6 & 3 & 1 & 0 \\ -7 & 0 & 0 & 0 \end{vmatrix}}{\begin{vmatrix} -1 & 2 & 1 & 1 \\ 3 & -1 & -3 & 0 \\ 3 & 3 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{vmatrix}} = \frac{\begin{vmatrix} -6 & -1 & -3 \\ 6 & 3 & 1 \\ -7 & 0 & 0 \end{vmatrix}}{\begin{vmatrix} 3 & -1 & -3 \\ 3 & 3 & 1 \\ 1 & 0 & 0 \end{vmatrix}} = \frac{-7\begin{vmatrix} -1 & -3 \\ 3 & 1 \end{vmatrix}}{\begin{vmatrix} -1 & -3 \\ 3 & 1 \end{vmatrix}} = -7$$

$$x_2 = \frac{\begin{vmatrix} -1 & 9 & 1 & 1 \\ 3 & -6 & -3 & 0 \\ 3 & 6 & 1 & 0 \\ 1 & -7 & 0 & 0 \end{vmatrix}}{\begin{vmatrix} A \end{vmatrix}} = \frac{\begin{vmatrix} 3 & -6 & -3 \\ 3 & 6 & 1 \\ 1 & -7 & 0 \end{vmatrix}}{8} = \frac{\begin{vmatrix} 0 & 15 & -3 \\ 0 & 27 & 1 \\ 1 & -7 & 0 \end{vmatrix}}{8} = \frac{\begin{vmatrix} 15 & -3 \\ 27 & 1 \end{vmatrix}}{8} = \frac{96}{8} = 12$$

$$x_3 = \frac{\begin{vmatrix} -1 & 2 & 9 & 1 \\ 3 & -1 & -6 & 0 \\ 3 & 3 & 6 & 0 \\ 1 & 0 & -7 & 0 \end{vmatrix}}{\begin{vmatrix} A \end{vmatrix}} = \frac{\begin{vmatrix} 3 & -1 & -6 \\ 3 & 3 & 6 \\ 1 & 0 & -7 \end{vmatrix}}{8} = \frac{\begin{vmatrix} 0 & -1 & 15 \\ 0 & 3 & 27 \\ 1 & 0 & -7 \end{vmatrix}}{8} = \frac{\begin{vmatrix} -1 & 15 \\ 3 & 27 \end{vmatrix}}{8} = -9$$

$$x_4 = \frac{\begin{vmatrix} -7 & 2 & 1 & 9 \\ 0 & 5 & 0 & 21 \\ 0 & 9 & 4 & 33 \\ 0 & 2 & 1 & 2 \end{vmatrix}}{\begin{vmatrix} -1 & 2 & 1 & 1 \\ 0 & 5 & 0 & 3 \\ 0 & 9 & 4 & 3 \\ 0 & 2 & 1 & 1 \end{vmatrix}} = \frac{\begin{vmatrix} 5 & 0 & 21 \\ 9 & 4 & 33 \\ 2 & 1 & 2 \end{vmatrix}}{\begin{vmatrix} 5 & 0 & 3 \\ 9 & 4 & 3 \\ 2 & 1 & 1 \end{vmatrix}} = \frac{\begin{vmatrix} 5 & 0 & 21 \\ 1 & 0 & 25 \\ 2 & 1 & 2 \end{vmatrix}}{\begin{vmatrix} 5 & 0 & 3 \\ 1 & 0 & -1 \\ 2 & 1 & 1 \end{vmatrix}} = \frac{\begin{vmatrix} 5 & 21 \\ 1 & 25 \end{vmatrix}}{\begin{vmatrix} 5 & 3 \\ 1 & -1 \end{vmatrix}} = -13$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -7 \\ 12 \\ -9 \\ -13 \end{bmatrix} \quad \#$$

4. Let A be the matrix in Exercise 3. Compute the second column of A^{-1} by using Cramer's rule to solve $A\mathbf{x} = \boldsymbol{e}_2$.

$$\begin{bmatrix} 2 & 1 & 2 \\ 0 & 3 & 4 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} | \\ x \\ | \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 4 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} | \\ x \\ | \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$x_1 = \frac{\begin{vmatrix} 0 & 0 & 0 \\ 1 & 3 & 4 \\ 0 & 1 & 2 \end{vmatrix}}{\begin{vmatrix} & A & \end{vmatrix}} = \frac{0}{|A|} = 0$$

$$x_2 = \frac{\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 1 & 0 & 2 \end{vmatrix}}{\begin{vmatrix} 1 & 0 & 0 \\ 0 & 3 & 4 \\ 1 & 1 & 2 \end{vmatrix}} = \frac{\begin{vmatrix} 1 & 4 \\ 0 & 2 \end{vmatrix}}{\begin{vmatrix} 3 & 4 \\ 1 & 2 \end{vmatrix}} = \frac{2}{2} = 1$$

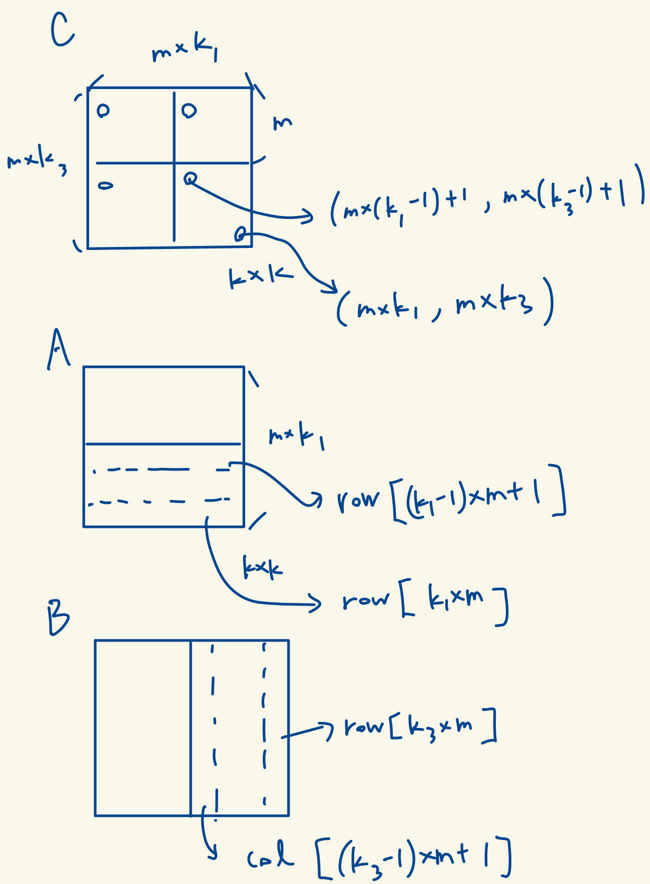
$$x_3 = \frac{\begin{vmatrix} 1 & 0 & 0 \\ 0 & 3 & 1 \\ 1 & 1 & 0 \end{vmatrix}}{\begin{vmatrix} & A & \end{vmatrix}} = \frac{\begin{vmatrix} 3 & 1 \\ 1 & 0 \end{vmatrix}}{\begin{vmatrix} 3 & 4 \\ 1 & 2 \end{vmatrix}} = -\frac{1}{2}$$

$$A^{-1} \cdot \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = A^{-1} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \mathrm{col}_2(A^{-1}) \text{ and } Ax = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow x = A^{-1} \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Say, $\mathrm{col}_2(A^{-1}) = x = \begin{bmatrix} 0 \\ 1 \\ -1/2 \end{bmatrix}$

程式作業

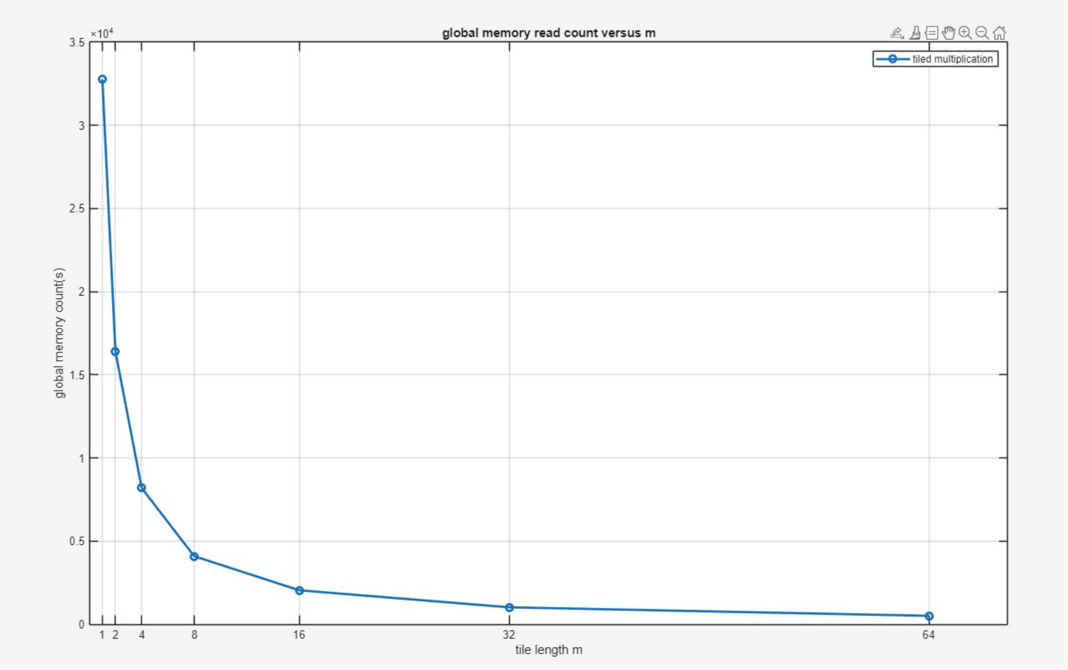
1-2 squence problem for tiled multiplication



In program, k_1 is replaced i and k_3 is replaced j

then we can get the index in each $m \times m$ tiled matrices.

1.3 graph

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% discussion part
% (1), for the question whether C_naive and C_tiled are accurate.
% From the numerical and plotting results, we can see both naive_multiply and tiled_multiply
% are extremely accurate with both rms = 0
% (2), about the interpretation about the count versus m plot, we can
% find that the counts decrease when length m doulbes. With this fact,
% I have a conclusion is that when the m is larger, the reading efficiency
% enhances. We can find this property with single tile as well. To
% calculate the read counts from global memory, we have m by m tile, counts
% by naive method is 2 (m*m). However, counts by tiled
% method is merely 2 * m = 2 * m since the data in shared memory are reused.
% the sum count number of two methods (compute whole nxn matrix)
% are 2 * m^2 * (k1 * k2) and 2 * m * (k1 * k2) --> rewrite as 2 * n^2
% and 2 * ( (n^2) / m), which is clear that tiled one is more efficient, and
% when m increase the counts being smaller.
% p.s. k1 = k2 = n/m
```

```
% p.s. extra thinking. Although tiled multiply is better, it also means it
% need bigger shared memory space to store those temporary data and
% in common, more efficent memory are more expensive and small, so it is still
% a good method to calculate but not always the best one depending on the
% situation.
```